11.11 How Fielders Catch

WORKSHEET

The question

STEP II 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q11

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Put the batsman at the origin and the fielder at distance D. If the ball is at (x, y), what is $\tan\theta$?

STAGE 1 The perfectly placed fielder

Covers: the stationary fielder

Ask yourself Put the batsman at the origin and the fielder at distance D. What is $\tan\theta$ when the ball is at (x, y)?

Pause and try Let the ball land at eye level at time T, where the fielder is. Write y and $D - x$ in terms of t and T. Can you factorise y so that something cancels?

STAGE 2 The running fielder

Covers: the running fielder

Ask yourself Now the fielder starts at distance D and runs towards the batsman at speed v. If she catches the ball at time T, what is D? Repeat the calculation.

Pause and try For the converse, suppose $\tan\theta$ rises at a constant rate c, so $\tan\theta = ct$. Clear the fraction and compare the coefficients of t and the constant terms. What does that tell you about D?

What's really going on

Optical acceleration cancellation

Chapman's rule is now called *optical acceleration cancellation*: the fielder moves so as to keep the "optical acceleration" $\frac{\mathrm{d}^2}{\mathrm{d}t^2}\tan\theta$ at zero. If the ball's image accelerates upwards, the ball will go over her head, so she backs away; if it decelerates, the ball will fall short, so she runs in (Figure 11.11.1 (in the full section online)). The strategy does not tell her where the ball will land, but it gets her there. It is a feedback control law, like the ones that steer guided missiles: missiles use *proportional navigation*, keeping the line-of-sight angle to the target constant, which is the sideways version of the same idea.

What fielders really do

Do real fielders use Chapman's rule? Experiments in the 1990s, with cameras mounted on the heads of baseball outfielders, gave mixed results. Michael McBeath and colleagues argued in *Science* in 1995 that, for balls hit to one side, fielders run along curved

paths that keep the ball's image moving in a straight line across their field of view (a "linear optical trajectory"), which also leads them to the catch. Later work has suggested that both strategies, and a mixture of them, are used, and that fielders often do not run at constant speed. A 2004 study found that dogs catching frisbees follow paths consistent with the same optical rules. Whatever the details, the remarkable conclusion stands: catching does not require prediction, only the right feedback.

Chapman's paper

Seville Chapman's "Catching a baseball", published in the *American Journal of Physics* in 1968, is three pages long and contains essentially the calculation in this question. He noted that the strategy works only without air resistance, and that a real baseball's trajectory is distorted by drag and spin, which would complicate things. Nevertheless, the idea that the brain uses simple invariants of the visual field to solve complicated control problems has become important in psychology and robotics.

Try this next

1. **Running backwards.** If the ball will land behind the fielder's starting position, show that she should run away from the batsman, and that the same rule applies.

2. **Not θ itself.** Show that, for a perfectly placed fielder, $\frac{\mathrm{d}\theta}{\mathrm{d}t}$ is not constant. Why is $\tan\theta$ the right quantity to watch?

3. **Eye level.** If the ball is struck from a height h below the fielder's eye level, is $\tan\theta$ still linear in t for a well-placed fielder? (Assume she catches it at eye level.)

4. **Air resistance.** With linear air resistance, the horizontal velocity decays exponentially. Is $\tan\theta$ still linear in t for a well-placed fielder?

Further reading

- **Seville Chapman, "Catching a baseball"**, *American Journal of Physics* 36 (1968).
- **Michael McBeath, Dennis Shaffer & Mary Kaiser, "How baseball outfielders determine where to run to catch fly balls"**, *Science* 268 (1995).
- **Dennis Shaffer et al., "How dogs navigate to catch Frisbees"**, *Psychological Science* 15 (2004).

The full guided solution is in Section 11.11:
https://beyondstep.keithmcnulty.org/ch11/fielders.html